Manos Adamopoulos

AI Engineer | LLM, Data & ML Systems

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Summary

Senior AI Engineer with 5+ years architecting and shipping reliable AI and ML systems on Cloud owning the full lifecycle from requirements through CI/CD, deployment, and monitoring. Two MSc degrees in Software and Engineering, **both with honors**. I have worked mostly with startups within ambiguous and fast-moving environments. Currently building **agent skills** and **RAG systems** at the core of GWI's AI transformation, and a member of the **AI Enablement team** defining agentic engineering standards across the organisation.

Professional Experience

Sr. AI Engineer | GWI | Oct 2023 – Present

Project: Lynx - Synthetic Survey Generation Agent (RAG)

- Architected and shipped in **3 months,** a production **AI orchestration** backend (Python, Django, DRF, Argo Workflows) that produces **synthetic survey data** from a user hypothesis. An high-end product that cuts survey generation from **a month to 5 minutes,** serving **~12K requests at peak hour**
- Deployed and operated **cloud-native infrastructure on GKE** via custom **Helm** charts and **Terraform/Terragrunt.**
- Built an **LLM synthetic-respondent engine** with **two-stage RAG** (OpenAI embeddings, cosine top-k) that grounds survey drafting in curated example questions and each simulated respondent's answers in real prior survey responses.

Project: Segmentation Integration Automation (SIA) - ML pipeline to predict custom audience segments on GWI's Core dataset

- Engineered and productionized an **event-driven ML post-processing microservice on GCP** (Cloud Run, Pub/Sub, Docker, Terraform) that rebalances weighted audience distributions for tens of thousands of respondents per run, guaranteeing convergence within 0.5% of market targets via a deterministic, parallelized iterative optimizer with early stopping.
- Designed a novel **LLM-agent optimizer** (**GPT-4o** function-calling loop, **3 custom domain tools**) that autonomously measures distribution divergence and reweights segment assignments; extended the engine to multilabel segmentation and achieved 100% ground-truth respondent retention, fixing a production-critical reassignment defect.

Platform and Company wide events

- Led adoption of engineering standards for the department as **DevOps Champion**, driving **AI Agentic engineering harness** adoption via the AI Enablement team and migrating department infrastructure into GWI's central **Terraform** repo with resource labelling that enabled per-product cost attribution
- **Drive AI literacy** across the organisation: speak at public discussions, run an **AI News Forum** that keeps teams informed on developments and encourages adoption

MLOps Engineer | Linked Business | Mar 2021 - Oct 2023

- Architected and built **two automated pipelines** for batch (Airflow-orchestrated for quarterly releases) and streamline (event-driven with scheduled pollers detecting source changes) data accordingly, replacing a manual data-retrieval process, saving **3 engineer-days per quarter** and moving data freshness from monthly to near real-time
- Built and deployed **ML-driven insight services on Vertex AI** (online real-time inference) and Cloud Run. Python FastAPI services retrieved and cleansed source data, which was preprocessed, deduplicated, and loaded into BigQuery. A calculation engine then turned the model outputs into lead-generation and onboarding insights for banks and sales teams.
- Productionized the platform with MLOps best practices: **Dockerized** services, **CI/CD** pipelines for automated testing and deployment, and a highly available **Kubernetes** cluster for zero-downtime releases and resilient data processing.

Data Scientist | Sociality co-op | Dec 2019- Mar 2021

- Built an end-to-end analytics pipeline on website activity data (Python, SQL, dbt, Airflow), using behavioral segmentation and lead-scoring models (scikit-learn, XGBoost) to surface high-potential leads and shape company strategy, with insights delivered through interactive dashboards (Power BI/Streamlit).
- Advised EU-funded, community-driven projects on data science implementation, turning local social data into actionable, evidence-based decisions

Education

- **MSc** in Software Development & AI - University of Piraeus 9.8/10
- **MSc** in Civil Engineering (5-year degree) - Democritus University of Thrace 7.04/10

Projects

Job Matcher - full-stack AI platform for job hunters. **Visit website**

A platform that helps job seekers describe roles in natural language, search listings from extracted preferences, and get concise, preference-aware explanations of why results fit.

- Sequential **multi-agent backend** (preference extractions and match advisor) with evidence-backed prompts from vector retrieval over API docs **(text-embedding-3-small, ChromaDB).**
- **Failure-tolerant RAG integration** (lazy service init, continue without context on errors).
- **React/TypeScript UI, FastAPI** services, **Docker** images and **Kubernetes** deployment.
- **GitHub Actions** CI builds backend/frontend Docker images on every push/PR to main.

Stack: Python (FastAPI) + React/TypeScript (Vite, Tailwind) + OpenAI + ChromaDB RAG, containerized with Docker and deployable on Kubernetes, with GitHub Actions CI

SkyScribe-CLI - multi-agent weather app

Python CLI that turns live forecast data into short, actionable weather briefs (temperature ranges, rain windows, wind, practical tips) for travelers and daily planning, using Open-Meteo APIs.

- **Open model + PEFT**: Fine-tuned **TinyLlama with LoRA** adapters (**Transformers** + **PEFT**)—train and ship small adapters instead of full weights.
- **Training/inference stack**: Supervised pairs with prompt-token label masking; optional 4-bit QLoRA for GPU-efficient training; adapter hot-load at inference via env config.

Stack: Python, Typer, httpx, **Hugging Face Transformers**, PEFT.

Skills

Languages: Python, SQL, TypeScript, JavaScript, HTML/CSS
Backend & APIs: FastAPI, Django REST Framework, REST API design, event-driven microservices (Pub/Sub), PostgreSQL, Redis
Cloud & DevOps: GCP (Cloud Run, GKE, Cloud Build, Artifact Registry, Secret Manager, GCS, BigQuery), Docker, Kubernetes/Helm, Argo Workflows, Terraform, GitHub Actions
Reliability & security: Secret Manager/SOPS, signed URLs, Sentry, structured logging, pytest, Playwright
ML/LLM: LightGBM, scikit-learn, SHAP, RAG, embeddings, ChromaDB, Hugging Face/PEFT
Frontend: React, Tailwind